

Signal Conduction

Note: In OneNote, set this worksheet as background. Use "Export" in the simulation and insert the image in the marked space.

Information text

The axon hillock forms the trigger zone of the neuron. Here, postsynaptic potentials are integrated spatially and temporally. If the membrane potential exceeds the threshold, an action potential is triggered. Information is not encoded by the height of a single action potential, but mainly by the frequency and pattern of successive action potentials.

The axon is specialised for rapid long-distance conduction. Its membrane contains voltage-gated ion channels, whose orderly opening and closing enables conduction of the action potential. In myelinated nerve fibres, glial cells surround sections of the axon with an electrically insulating myelin sheath. As a result, ion flow mainly occurs at the nodes of Ranvier. The excitation is therefore conducted saltatorily from node to node. This form of conduction increases speed and at the same time reduces energy expenditure, because ion gradients have to be restored especially intensively only at small membrane sections. An unmyelinated axon has no such insulating covering. Therefore, depolarisation must be triggered again at every neighbouring membrane section. Continuous conduction is therefore slower and more energy-consuming because much larger membrane areas are directly involved in changes in membrane potential.

At the molecular level, conduction is based on the regulated opening and closing of voltage-gated sodium and potassium channels. Local currents depolarise the neighbouring membrane section to threshold, so that the action potential moves along the axon. The myelin sheath increases the electrical resistance of the membrane and at the same time reduces its effective membrane capacitance. As a result, local currents lose less charge and reach the next node of Ranvier particularly quickly. In addition, voltage-gated ion channels are present in high density mainly at the nodes of Ranvier. When the threshold is reached there, sodium channels open again and the action potential spreads further. Shortly afterwards, sodium channels inactivate while voltage-gated potassium channels open. Potassium ions flow out of the cell; repolarisation and often temporary hyperpolarisation occur. This temporal sequence is crucial for directed conduction because the area that has just been excited is initially absolutely or relatively refractory.

Task 1: Comparing saltatory and continuous conduction

Compare conduction in the upper and lower axon. Describe how you can tell that excitation is conducted saltatorily in the upper axon and continuously in the lower axon, and name two advantages of saltatory conduction.

Task 2: Explaining conduction

Describe saltatory conduction at the molecular level. Include local currents, voltage-gated sodium and potassium channels and the role of the nodes of Ranvier.

Task 3: Model experiment on conduction

Build two models of signal conduction in neurons: one for continuous conduction and one for saltatory conduction. Use the dominoes and straws provided and photograph your models.

Image:

*Model of continuous
conduction*

Image:

Model of saltatory conduction